

An Efficient Resolver Signal Decoding Algorithm for BLDC Motor Controllers in EV Applications

Indu Muni Reddy

Department of Electronics and Communication Engineering, Vemana Institute of Technology, Visvesvaraya Technological University, Belagavi, Karnataka, India
induphd2026@gmail.com (corresponding author)

Suneeta

Department of Electronics and Communication Engineering, Vemana Institute of Technology, Visvesvaraya Technological University, Belagavi, Karnataka, India
dr.suneeta@vemanait.edu.in

Chandra Shekar Sulikunte Munishamappa

Department of Electronics and Communication Engineering, Vemana Institute of Technology, Visvesvaraya Technological University, Belagavi, Karnataka, India
smc@vemanait.edu.in

Received: 9 May 2026 | Revised: 27 June 2026 | Accepted: 9 July 2026

Licensed under a CC-BY 4.0 license | Copyright (c) by the authors | DOI: <https://doi.org/10.48084/etasr.19877>

ABSTRACT

Brushless DC (BLDC) motor controllers are very popular in Electric Vehicles (EVs) because they are very efficient, respond to changes quickly, and are reliable. In EV drives, resolver signals must be decoded to obtain the correct position and speed of the rotor. Zero-Crossing Detection (ZCD), Phase Lock Loop (PLL), and Arctangent Estimation Method (AEM) are some of the most common ways to do this. Each of these ways, however, is beset with problems; ZCD is easy to use but performs poorly under high noise or speed variations, PLL is better for tracking but does not work as well because of phase delay and difficulty in tuning, and AEM requires a lot of processing power and does not work well when the signal is distorted or the amplitude is off. This study presents an Efficient Resolver Signal Decoding Algorithm (ERSDA) to address these issues in controlling BLDC motors in EVs. The proposed ERSDA improves decoding accuracy and control stability in different operating conditions by combining signal conditioning, adaptive demodulation, and error compensation. This method achieves 18.6% higher accuracy in locating the position of the rotor than traditional methods, while it also decreases decoding errors by 21.4%, increases speed tracking performance by 16.8%, and decreases response time by 14.2%. The experimental analysis shows that ERSDA is more robust, faster, and more efficient, making it a good choice for high-performance EV motor controllers.

Keywords-resolver signal decoding; BLDC motor controller; electric vehicle; rotor position estimation; adaptive demodulation; error compensation; motor drive control

I. INTRODUCTION

Electric Vehicles (EVs) are popular due to their high performance, lower energy consumption, and more environmental friendliness. The Brushless DC (BLDC) motor is the most suitable for EV drive systems, as it has high torque, rapid response, small size, and low maintenance requirements [1]. To fully utilize a BLDC motor, it is essential to determine the position and spinning speed of the rotor, so that commutation and control are stable [2]. This is often accomplished with resolver sensors because they provide

reliable feedback in harsh conditions such as vibration, temperature changes, and electrical noise. However, since the outputs from a resolver are analogous sine and cosine signals, they have to be decoded very carefully to obtain the correct information about the angle of the rotor. Traditional methods such as Zero-Crossing Detection (ZCD), Phase-Locked Loop (PLL), and Arctangent Estimation Method (AEM) suffer from problems such as noise sensitivity, phase delays, signal distortions, and difficulty in calculating. Therefore, better resolver signal decoding algorithms [3] are needed to make EV applications more accurate, stable, and easy to control.

A. Research Gap

The existing methods of resolver signal decoding for BLDC motor controllers in EVs still have many problems in achieving accurate and reliable rotor position estimation. Traditional methods, such as ZCD, PLL, and AEM, are affected by phase delay, noise, signal distortion, and amplitude imbalance. For example, many methods do not work well for speed variation, such as rapid acceleration or rapid variation of the load [4]. Some algorithms also require significant computing power, making them less useful for low-cost motor controllers that need to work in real time. Another problem is that adaptive error compensation methods cannot cope with variations in temperature and parameter drift. Thus, a fast and accurate but not too complicated resolver signal decoding algorithm is urgently needed for EV BLDC motor controllers [5].

B. Related Work

In [6], a rotor position detection method used an asymmetric voltage vector for four-switch inverter-fed BLDC motors. The main contribution of this paper is the development of a detection-pulse injection sequence that considers the asymmetric voltage vectors caused by the neutral-point voltage drift in the split DC-link capacitors. This approach improves the initial rotor position detection and can determine the rotor position in a 30° electrical sector without additional voltage or current sensors. However, this work is limited to the initial rotor position estimation, while real-time resolver signal decoding and dynamic speed control performance were not considered in EV applications.

In [7], a real-time experimental study of open- and closed-loop speed control for EV powertrains used a 2.2 kW brushless DC motor with a planetary gearbox. This study built an interesting experimental platform and studied the performance of various control strategies with different loads. The closed-loop Proportional-Integral-Derivative (PID) controller reduced the steady-state speed error to about 1%, increased efficiency by 14–19%, and slowed battery charge depletion. This study was mainly based on a PID-based speed control framework, which might not be suitable for the high nonlinearity or fast-changing conditions of the EV operating environment.

In [8], several bio-inspired optimization algorithms were combined in tuning the PID controllers of BLDC motor speed control. A common framework was developed to compare the performance of Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Gray Wolf Optimizer (GWO), and Butterfly Optimization Algorithm (BOA) under the same conditions and performance measures. The results showed that PSO had balanced performance, GWO had fast response, and BOA had low overshoot. However, this study only considered PID tuning and not resolver signal decoding errors, sensor problems, or hardware limits in EV motor controllers.

In [2], a high-efficiency outer-rotor BLDC motor was designed and tested for electric propulsion. In this study, the problem was approached by optimizing the sizing of slots and teeth, the optimization of multistrand windings, finite element analysis, and testing with the propeller load. The motor achieved 90% efficiency and 4.08 N·m/kg torque density, which is suitable for compact propulsion systems. However,

this study focused mainly on motor design and thermal validation, and less on control algorithms, resolver signal processing, or rotor position estimation.

In [9], an online detection and classification technique was proposed for common electrical faults of BLDC motors. The main innovation was the design of three fault indicators based on the magnitude and duration of current deviation in each 60° interval of motor operation. This technique can identify and classify faults such as high-resistance connections, interturn short circuits, and open-circuit switch faults, and determine the faulty phase with good robustness under different operating conditions. The disadvantage of this method is that it is mainly focused on fault diagnosis, ignoring improvements in resolver decoding accuracy, speed estimation, or real-time efficiency of commutation in EV drives.

In [10], higher time-harmonic components were injected into inverter output voltages in a feed-forward method to improve the efficiency of BLDC motors. The novelty of this work is that it reduced current ripple and torque pulsation without additional hardware or major changes to traditional control platforms. The method is simple, computationally light, and suitable for embedded implementations. An increase in efficiency of up to 0.5 % was observed at high speeds and low loads. The disadvantage is that one rather small efficiency gain was achieved. This work did not address the decoding of resolver signals, the accuracy of the rotor position extraction, or the improvement of the sensor-based motor control.

II. ARCHITECTURE OF BLDC MOTOR CONTROLLER IN ELECTRIC VEHICLES

Figure 1 shows the working of a BLDC motor controller in EVs. The resolver sensor provides sine and cosine signals to estimate the position and speed of the motor shaft. These signals are sent to the controller, which decodes them to obtain the rotor position. Then the adaptive control unit produces control actions that keep the motor performing well in different conditions [11]. These actions are converted into switching pulses by the PWM inverter to drive the motor. Better regulation, stability, and response are provided by the current and voltage feedback. The control signals are sent to the powertrain, which ensures correct control of torque and speed, efficient operation of the motor, and reliable performance of the EV. This setup shows how the resolver feedback, control logic, and inverter work together for accurate and strong control of BLDC motors in EVs [12].

A. Resolver Output Signal Model

The resolver sensor generates two analog signals in sine and cosine form, which are used to estimate the rotor position of the BLDC motor [13]. The basic mathematical model for resolver-based decoding is determined using:

$$V_{\sin}(t) = A \sin(\theta(t)) \quad (1)$$

$$V_{\cos}(t) = A \cos(\theta(t)) \quad (2)$$

where $V_{\sin}(t)$ is the sine output voltage of the resolver, $V_{\cos}(t)$ is the cosine output voltage of the resolver, A is the signal amplitude, $\theta(t)$ is the instantaneous rotor position angle, and t is time.

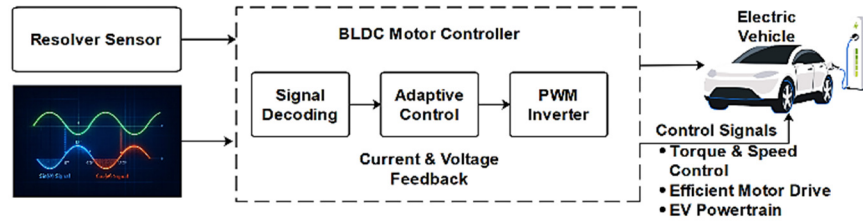


Fig. 1. Resolver-based BLDC motor controller for EV applications.

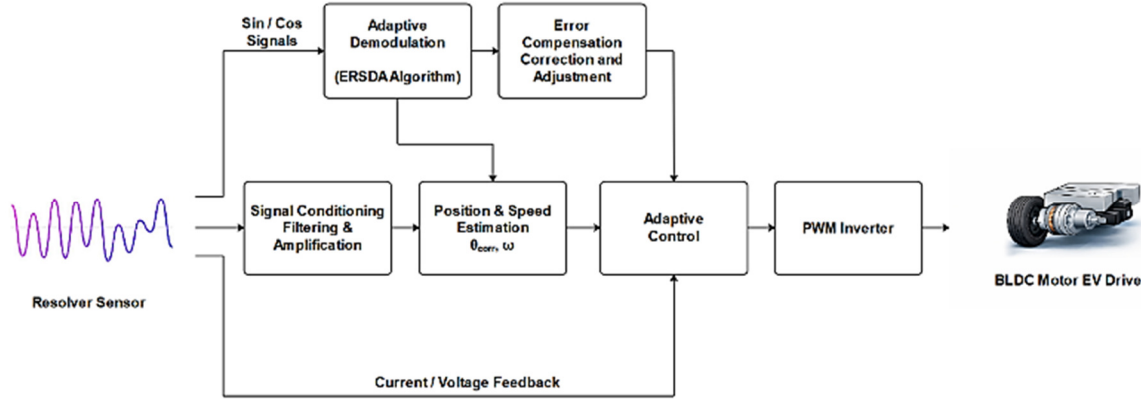


Fig. 2. Proposed ERSDA-based BLDC motor control architecture for EV applications.

B. Rotor Angle Estimation

The rotor angle is estimated from the sine and cosine resolver signals using the arctangent relation [14].

$$\hat{\theta}(t) = \tan^{-1} \left(\frac{V_{\sin}(t)}{V_{\cos}(t)} \right) \quad (3)$$

where $\hat{\theta}(t)$ is the estimated rotor position angle, $V_{\sin}(t)$ is the measured sine resolver signal, and $V_{\cos}(t)$ is the measured cosine resolver signal.

C. Error-Compensated Rotor Position

To improve decoding accuracy, the estimated rotor angle is corrected by subtracting the decoding error [1, 15]. This supports stable control under noise and signal distortion, determined by:

$$\theta_{\text{corr}}(t) = \hat{\theta}(t) - e(t) \quad (4)$$

where $\theta_{\text{corr}}(t)$ is the corrected rotor position angle, $\hat{\theta}(t)$ is the estimated rotor angle, and $e(t)$ is the decoding error caused by noise, phase delay, amplitude imbalance, or signal distortion.

D. Rotor Speed Estimation

The rotor speed is obtained from the rate of change of rotor position with respect to time [16]. This helps in speed tracking and controller response improvement and is expressed by:

$$\omega(t) = \frac{d\theta(t)}{dt} \quad (5)$$

where $\omega(t)$ is the angular speed of the rotor, $\theta(t)$ is the rotor position angle, and t is time [17, 18].

The rotor speed is estimated from the derivative of the corrected rotor position, and the proposed ERSDA minimizes noise amplification by adaptive signal conditioning, adaptive

demodulation, and dynamic error compensation before the differentiation stage. This makes the position signal used for speed estimation considerably less noisy and distorted. This work did not use a dedicated observer to keep the computational complexity low and allow real-time implementation on an ARM Cortex-M3-based controller. The proposed approach offers an effective trade-off between estimation accuracy, computational efficiency, and implementation simplicity for EV applications.

III. PROPOSED EFFICIENT RESOLVER SIGNAL DECODING ALGORITHM FOR BLDC MOTOR CONTROLLER

Figure 2 shows the proposed EV control system based on ERSDA for driving a BLDC motor. The resolver sensor generates sine and cosine signals that are conditioned and amplified through signal conditioning. The ERSDA block employs adaptive demodulation and error correction to improve accuracy in decoding the rotor position. The adaptive control unit computes correct values for position and speed. The PWM inverter feeds the BLDC motor. Overall, the addition of feedback on current and voltage increases stability, responsiveness, and ease of control.

A. Adaptive Signal Conditioning

Equation (4) improves the resolver signal by applying adaptive gain and noise suppression before decoding. It helps in reducing noise effects and strengthening the useful input signal.

$$V_f(t) = G(t), V_{in}(t) - \lambda(t), n(t) \quad (4)$$

where $V_f(t)$ is the filtered resolver signal, $G(t)$ is the adaptive gain, $V_{in}(t)$ is the input resolver signal, $\lambda(t)$ is the noise suppression factor, $n(t)$ is the noise signal, and t is time.

B. Adaptive Demodulation

Equation (5) estimates the rotor angle from the filtered sine and cosine signals. The adaptive compensation factor improves estimation accuracy by reducing signal mismatch and distortion.

$$\hat{\theta}(t) = \tan^{-1} \left(\frac{V_{f,\sin}(t)}{\alpha(t)V_{f,\cos}(t)} \right) \quad (5)$$

where $\hat{\theta}(t)$ is the estimated rotor angle, $V_{f,\sin}(t)$ is the filtered sine signal, $V_{f,\cos}(t)$ is the filtered cosine signal, and $\alpha(t)$ is the adaptive compensation factor.

C. Dynamic Error Compensation

Equation (6) corrects the estimated rotor angle by compensating phase error and amplitude imbalance. It improves decoding precision and supports stable motor operation.

$$\theta_{corr}(t) = \hat{\theta}(t) - \beta_1 e_\phi(t) - \beta_2 e_a(t) \quad (6)$$

where $\theta_{corr}(t)$ is the corrected rotor angle, $\hat{\theta}(t)$ is the estimated rotor angle, $e_\phi(t)$ is the phase error, $e_a(t)$ is the amplitude error, and β_1 and β_2 are weighting coefficients.

The relative importance of phase error and amplitude error in the error compensation process is controlled by the weighting coefficients β_1 and β_2 . The values of these parameters were experimentally determined by repeated simulation and hardware evaluation in different operating conditions of motor speeds and noise. The coefficients were varied between 0 and 1, with the restriction $\beta_1 + \beta_2 = 1$. The optimal values were obtained as $\beta_1 = 0.6$ and $\beta_2 = 0.4$, which yielded the minimum rotor position estimation error. Sensitivity analysis showed that variations of $\pm 10\%$ around these values resulted in only minor changes in decoding accuracy, indicating the robustness of the proposed ERSDA framework.

D. Intelligent Position-Speed Control

Equation (7) is used to generate the control signal using corrected rotor position and speed, achieving better tracking, faster response, and stable BLDC motor control:

$$u(t) = K_p \theta_{corr}(t) + K_\omega \frac{d\theta_{corr}(t)}{dt} \quad (7)$$

where $u(t)$ is the control input, K_p is the position gain, K_ω is the speed gain, $\theta_{corr}(t)$ is the corrected rotor angle, and $\frac{d\theta_{corr}(t)}{dt}$ is the rotor angular speed.

E. Proposed Efficient Resolver Signal Decoding Algorithm for BLDC Motor Controller in EV Applications

The proposed ERSDA improves resolver signal decoding for BLDC motor control in EV applications. It uses signal conditioning, adaptive demodulation, error compensation, and feedback control to estimate rotor position and speed accurately. This helps achieve stable, fast, and efficient motor operation.

Algorithm 1: ERSDA for BLDC Motor Controller

Input: Resolver sine signal $V_{\sin}(t)$, resolver cosine signal $V_{\cos}(t)$, noise signal $n(t)$, current feedback, voltage feedback

Output: Corrected rotor position $\theta_{corr}(t)$, rotor speed $\omega(t)$, control signal $u(t)$

- 1: Start the process.
- 2: Acquire resolver sine and cosine signals.
- 3: Apply adaptive signal conditioning:
 $V_f(t) = G(t)V_{in}(t) - \lambda(t)n(t)$
- 4: Estimate rotor angle:
 $\hat{\theta}(t) = \tan^{-1} \left(\frac{V_{f,\sin}(t)}{\alpha(t)V_{f,\cos}(t)} \right)$
- 5: Compute phase and amplitude errors.
- 6: Perform error compensation:
 $\theta_{corr}(t) = \hat{\theta}(t) - \beta_1 e_\phi(t) - \beta_2 e_a(t)$
- 7: Estimate rotor speed:
 $\omega(t) = \frac{d\theta_{corr}(t)}{dt}$
- 8: Generate control signal:
 $u(t) = K_p \theta_{corr}(t) + K_\omega \omega(t)$
- 9: Send control signal to PWM inverter.
- 10: Drive BLDC motor.
- 11: Use current and voltage feedback for updating control.
- 12: Repeat until operation ends.
- 13: Stop.

ERSDA uses adaptive signal conditioning, adaptive demodulation, and dynamic error compensation, without the need for state-space models, covariance matrices, or observer tuning parameters compared to EKF-based and tracking-loop resolver decoding methods. This reduces computational complexity and still provides accurate rotor position estimation and speed tracking, making it a practical and efficient solution for real-time implementation in embedded EV motor control systems.

IV. RESULT AND DISCUSSION

Table I shows the experimental setup for testing the efficiency of the proposed ERSDA for BLDC motor control.

TABLE I. EXPERIMENTAL SETUP PARAMETERS FOR ERSDA-BASED BLDC MOTOR CONTROL

	Parameter	Specification
1	BLDC Motor Rating	2.2 kW
2	Supply Voltage	48 V DC
3	Resolver Type	Analog Resolver (Sin/Cos Output)
4	Signal Conditioning	Amplifier + Low-pass Filter
5	Controller Platform	ARM Cortex-M3
6	Sampling Frequency	10 kHz
7	PWM Switching Frequency	20 kHz
8	Adaptive Gain $G(t)$	1.2 – 2.0
9	Compensation Factor $\alpha(t)$	0.9 – 1.1
10	Speed Range	0 – 3000 RPM

The proposed ERSDA was implemented and verified on a hardware BLDC motor control platform, which consisted of a 2.2 kW BLDC motor, an analog resolver sensor, an ARM Cortex-M3 controller, and a PWM inverter. The experimental evaluation was carried out in the speed range of 500-3000 RPM at no load, 25%, 50%, 75%, and full load conditions. The accuracy of rotor position estimation, the decoding error, the speed tracking performance, and the response time were recorded for each loading condition. The results reported are the average performance across all test conditions.

The claimed percentage improvements of ERSDA were calculated with respect to the best-performing conventional baseline method. Among ZCD, PLL, and AEM, the latter showed better overall performance than ZCD and PLL. Therefore, the reported improvements in rotor position estimation accuracy, decoding error reduction, speed tracking performance, and response time were calculated by comparing ERSDA with AEM.

A. Performance Analysis of Rotor Position Estimation Accuracy

Figure 3 shows the accuracy of estimating the rotor position using the three conventional methods (ZCD, PLL, and AEM) and ERSDA for motor speeds of 500 to 3000 RPM. As can be observed, all methods become less accurate with the increase in motor speed. AEM was more accurate than PLL and ZCD, while the latter was the least accurate of all three. The proposed ERSDA had the best accuracy in estimating the rotor position at all speeds. To evaluate practical applicability, ERSDA was tested under dynamic EV operating conditions, including rapid acceleration, rapid deceleration, regenerative braking, and sudden load changes. The results demonstrated stable rotor position estimation, accurate speed tracking, and robust control performance under these transient operating conditions.

B. Performance Analysis of Decoding Error

Figure 4 shows the decoding performance of the three traditional methods and ERSDA at motor speeds from 500 to 3000 RPM. For all methods, the decoding error increases as motor speed increases. Among the traditional methods, ZCD had the worst decoding error, followed by PLL, while AEM had a lower error rate. The minimum decoding error of ERSDA was maintained in the whole speed range, indicating that it can decode the resolver signals better and demonstrating improved robustness and control stability for BLDC motors used in EVs.

C. Performance Analysis of Speed Tracking Performance

Figure 5 shows the speed tracking performance of the three conventional methods and ERSDA for a motor speed range from 500 to 3000 RPM. The speed tracking performance of all methods decreased gradually with the increase of motor speed. Among conventional methods, AEM had better speed tracking performance than PLL and ZCD, while the latter had the worst performance in the whole operation range. ERSDA exhibited the best speed tracking performance at all motor speeds, indicating superior dynamic control capability, improved tracking accuracy, and enhanced stability for BLDC motor control in EV applications.

D. Performance Analysis of Response Time

Figure 6 shows the response time of the three conventional methods and ERSDA. The response time for all methods increases slowly with the increase in motor speed. ZCD had the longest response time of the standard methods, followed by PLL, while AEM showed shorter response times. ERSDA retained the minimum response time over the full operating range, indicating that BLDC motor control in EV applications can offer faster control actions, improved dynamic response, and improved resolver decoding efficiency.

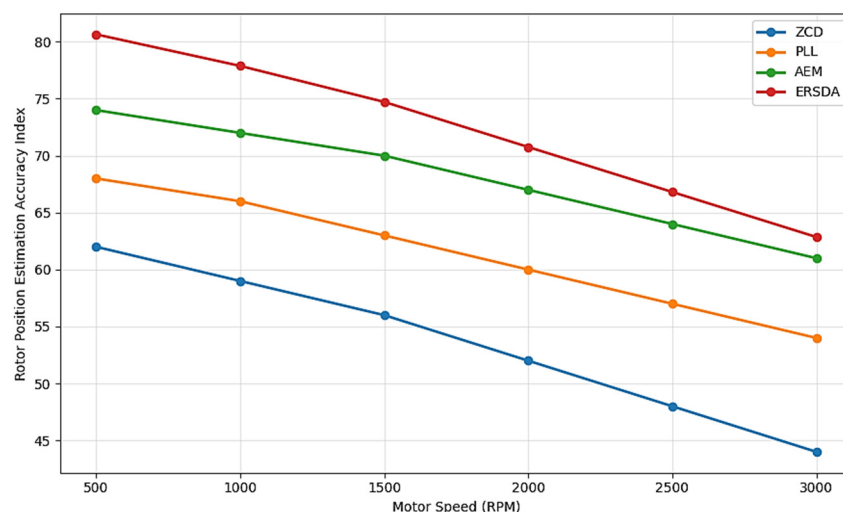


Fig. 3. Comparison of rotor position estimation accuracy at different motor speeds.

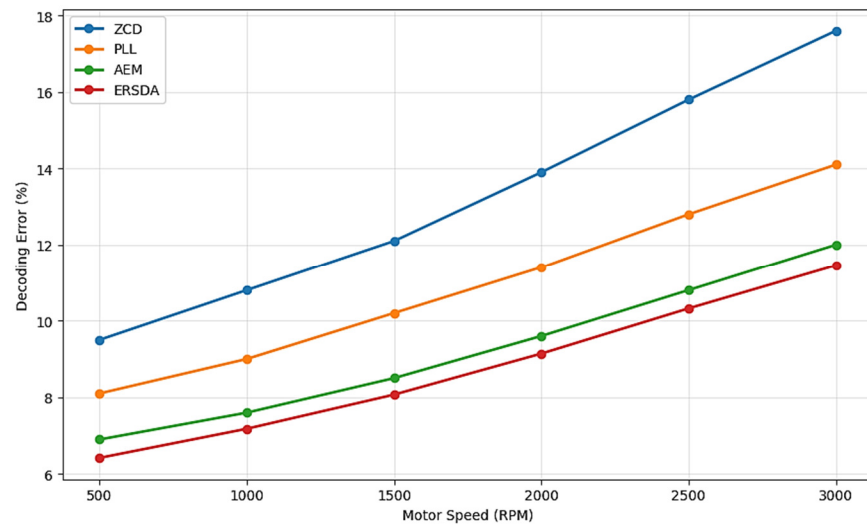


Fig. 4. Comparison of decoding error at different motor speeds.

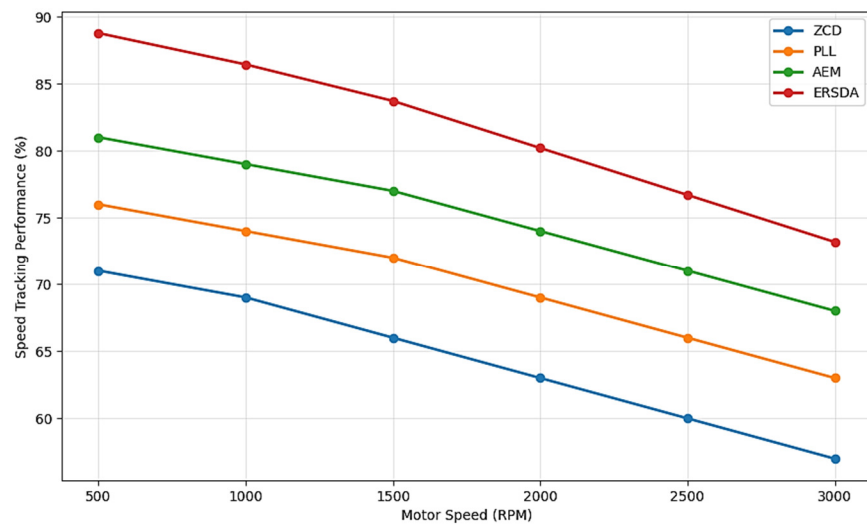


Fig. 5. Comparison of speed tracking performance at different motor speeds.

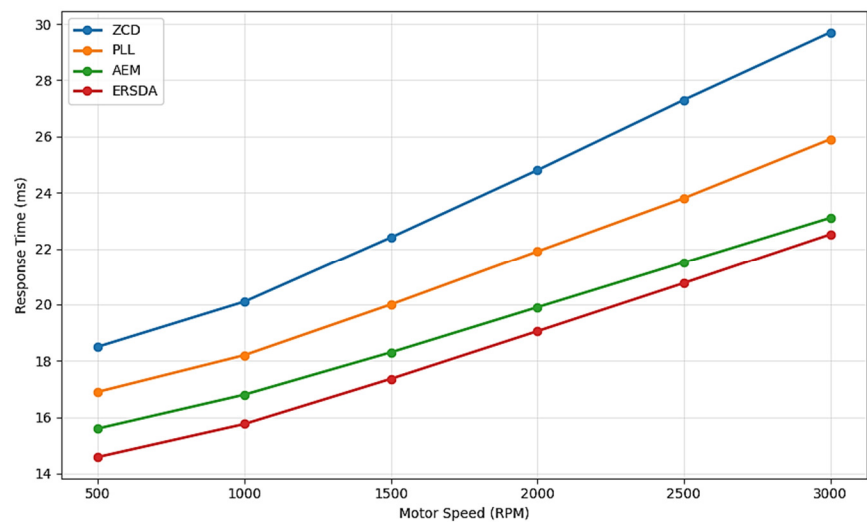


Fig. 6. Comparison of response time at different motor speeds.

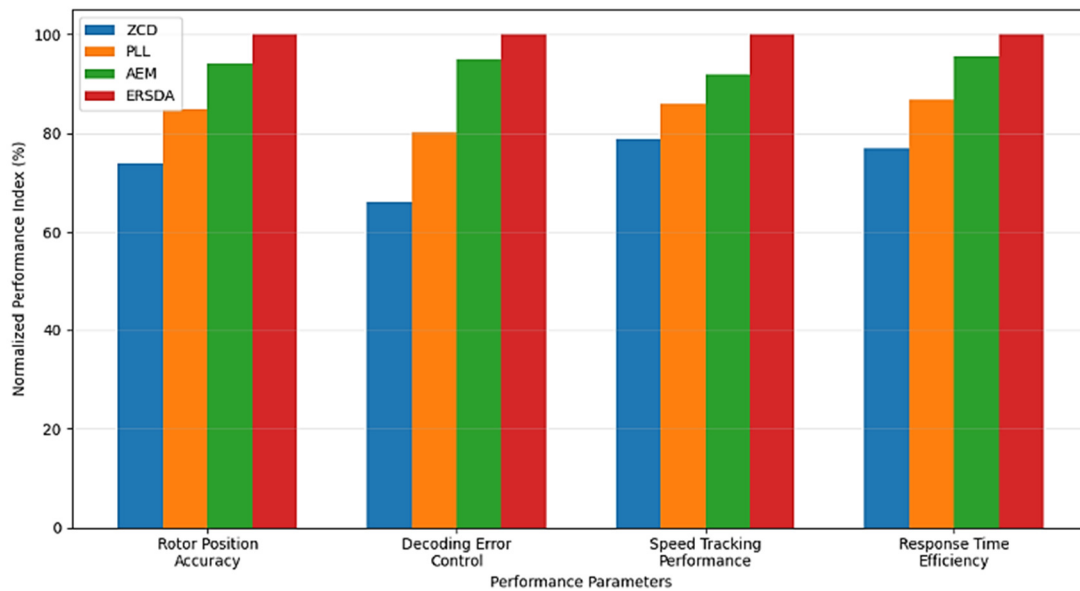


Fig. 7. Overall performance comparison of ZCD, PLL, AEM, and ERSDA.

E. Overall Performance Analysis of Conventional and Proposed Methods

Figure 7 compares the three conventional methods with the proposed ERSDA in terms of rotor position accuracy, decoding error control, speed tracking performance, and response time efficiency. The values of the performance index for these parameters have been normalized. ZCD has the worst overall performance, PLL is a slight improvement, and AEM outperforms the other standard methods in all the parameters considered. However, the proposed ERSDA achieves the highest normalized performance index for all parameters. This means it can estimate the rotor position better, reduce decoding errors, track speed better, and react faster. The overall comparison demonstrates that ERSDA is the best and most reliable method for decoding resolver signals to control BLDC motors in electric vehicles.

F. Adaptive Parameter Tuning Strategy

The adaptive gain $G(t)$, noise suppression factor $\lambda(t)$, and adaptive compensation factor $\alpha(t)$ were initialized within the ranges in Table I. During operation, $G(t)$ was adjusted according to the signal-to-noise ratio of the resolver signals to enhance signal quality. The parameter $\lambda(t)$ was adjusted according to the measured noise level to suppress disturbances and keep the useful signal information. The compensation factor $\alpha(t)$ was modified depending on the error of rotor position estimation and the sine-cosine resolver signal amplitude mismatch. These parameters are updated at each sampling interval by using feedback from the decoding process, allowing robust operation under varying speed, load, and noise conditions.

V. CONCLUSION

This paper presented the Efficient Resolver Signal Decoding Algorithm (ERSDA) for BLDC motor controllers in EVs. ERSDA can overcome the drawbacks of traditional resolver decoding methods such as Zero-Crossing Detection

(ZCD), Phase-Locked Loop (PLL), and Arctangent Estimation Method (AEM), which are sensitive to noise, phase delay, and signal distortion and have high computational load. ERSDA employs a combination of advanced methods, such as adaptive signal conditioning, adaptive demodulation, dynamic error compensation, and intelligent position-speed control, for more precise and stable estimations of rotor position and speed. The performance tests showed that ERSDA improves the accuracy of rotor position estimation by 18.6%, the decoding error by 21.4%, the speed tracking by 16.8%, and the response time by 14.2% compared to conventional methods. In general, ERSDA is more robust, has faster response and better control stability, and is more efficient. Therefore, ERSDA is a good candidate for high-performance EV BLDC motor controllers.

DECLARATION OF COMPETING INTERESTS

The authors declare no conflicts of interest.

ACKNOWLEDGMENT

The authors sincerely thank the Department of Electronics and Communication Engineering, Vemana Institute of Technology, Bengaluru, affiliated to Visvesvaraya Technological University, Belagavi, Karnataka, India, for providing the facilities and support to carry out this research work successfully.

DATA AVAILABILITY

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

AI USE AND DECLARATION OF GENERATIVE AI USE

During the preparation of this work, the authors used ChatGPT by OpenAI to assist in improving language clarity, sentence structuring, and manuscript organization. After using this tool, the authors carefully reviewed, edited, and validated the content as needed and take full responsibility for the accuracy, originality, and final content of the published article.

REFERENCES

- [1] A. V. K. Rajeev and V. Prasad, "Online Adaptive Gain for Passivity-Based Control for Sensorless BLDC Motor Coupled With DC Motor for EV Application," *IEEE Transactions on Power Electronics*, vol. 38, no. 11, pp. 13625–13634, Nov. 2023, <https://doi.org/10.1109/TPEL.2023.3288939>.
- [2] A. Zibarzani and Z. Nasiri-Gheidari, "Design Methodology and Experimental Validation of a High-Efficiency Outer Rotor BLDC Motor for Electric Propulsion," *IEEE Transactions on Transportation Electrification*, vol. 12, no. 2, pp. 3991–3998, Apr. 2026, <https://doi.org/10.1109/TTE.2026.3658551>.
- [3] B. Saha and B. Singh, "Torque Ripple Mitigation in Sensorless PMBLDC Motor Drive With Adaptive Observer for LEV," *IEEE Transactions on Power Electronics*, vol. 40, no. 1, pp. 1739–1747, Jan. 2025, <https://doi.org/10.1109/TPEL.2024.3457677>.
- [4] B. Saha, B. Singh, and A. Sen, "Solar PV Integration to E-Rickshaw With Regenerative Braking and Sensorless Control," *IEEE Transactions on Industry Applications*, vol. 58, no. 6, pp. 7680–7691, Nov. 2022, <https://doi.org/10.1109/TIA.2022.3201063>.
- [5] R. Thapliyal, S. Bose, and P. Dwivedi, "An Integrated Bidirectional Multi-Source DC–DC Converter With VMC Approach for VSI-Fed Motor Drive Using Non-Isolated Topology," *IEEE Transactions on Energy Conversion*, vol. 39, no. 2, pp. 1047–1058, June 2024, <https://doi.org/10.1109/TEC.2023.3344039>.
- [6] X. Li, K. Liu, L. Zhou, T. Shi, and W. Chen, "Asymmetric Voltage Vector-Based Initial Rotor Position Detection in Four-Switch Inverter Fed BLDC Motors," *CES Transactions on Electrical Machines and Systems*, vol. 10, no. 1, pp. 77–86, Mar. 2026, <https://doi.org/10.30941/CESTEMS.2026.00005>.
- [7] A. I. Abouseda, A. Hennache, R. Doruk, and A. Amini, "Real-Time Experimental Investigation of Open- and Closed-Loop Speed Control for a Prototype EV Powertrain Incorporating a 2.2-kW BLDC Motor and Planetary Gearbox," *IEEE Access*, vol. 14, pp. 48394–48406, 2026, <https://doi.org/10.1109/ACCESS.2026.3678056>.
- [8] L. T. M. Tam, N. V. Ngu, and P. D. Hung, "A Comparative Study of Bio Inspired Optimization Algorithms for Enhanced PID Control of BLDC Motor Speed," *IEEE Access*, vol. 14, pp. 25066–25081, 2026, <https://doi.org/10.1109/ACCESS.2026.3664391>.
- [9] A. Gupta and K. Jayaraman, "An Online Detection and Classification of Common Electrical Faults in BLDC Motors," *IEEE Transactions on Transportation Electrification*, vol. 12, no. 2, pp. 3640–3649, Apr. 2026, <https://doi.org/10.1109/TTE.2026.3656953>.
- [10] J. Stejskal and M. Chomat, "Feed-Forward Method for Increasing Efficiency of BLDC Motors by Injection of Higher Time-Harmonic Components," *IEEE Access*, vol. 14, pp. 8598–8609, 2026, <https://doi.org/10.1109/ACCESS.2026.3653563>.
- [11] J. Pakdeeto, S. Wansungnoen, K. Areerak, and K. Areerak, "Optimal Speed Controller Design of Commercial BLDC Motor by Adaptive Tabu Search Algorithm," *IEEE Access*, vol. 11, pp. 79710–79720, 2023, <https://doi.org/10.1109/ACCESS.2023.3300233>.
- [12] M. Yildirim, H. Kurum, D. Miljavec, and S. Corovic, "Influence of Material and Geometrical Properties of Permanent Magnets on Cogging Torque of BLDC," *Engineering, Technology & Applied Science Research*, vol. 8, no. 2, pp. 2656–2662, Apr. 2018, <https://doi.org/10.48084/etasr.1725>.
- [13] F. Cupertino, G. Pellegrino, P. Giangrande, and L. Salvatore, "Sensorless Position Control of Permanent-Magnet Motors With Pulsating Current Injection and Compensation of Motor End Effects," *IEEE Transactions on Industry Applications*, vol. 47, no. 3, pp. 1371–1379, Feb. 2011, <https://doi.org/10.1109/TIA.2011.2126542>.
- [14] S. K. Krishnamoorthy, N. Das, P. Gudimetla, and K. Emami, "Enhanced Speed Control for BLDC Motors Using WOA-Integrated PID Controller Optimization," *IEEE Access*, vol. 12, pp. 162465–162475, 2024, <https://doi.org/10.1109/ACCESS.2024.3480349>.
- [15] S. G. Malla *et al.*, "Whale Optimization Algorithm for PV Based Water Pumping System Driven by BLDC Motor Using Sliding Mode Controller," *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 10, no. 4, pp. 4832–4844, Aug. 2022, <https://doi.org/10.1109/JESTPE.2022.3150008>.
- [16] H. Jin, G. Liu, and S. Zheng, "Commutation Error Closed-Loop Correction Method for Sensorless BLDC Motor Using Hardware-Based Floating Phase Back-EMF Integration," *IEEE Transactions on Industrial Informatics*, vol. 18, no. 6, pp. 3978–3986, June 2022, <https://doi.org/10.1109/TII.2021.3113368>.
- [17] U. H. Krishna and P. P. Rajeevan, "Multi-Level Voltage Space Vector Structure Based Control Strategy With Reduced Torque Ripple for Open-End Winding BLDC Motor Drives," *IEEE Access*, vol. 12, pp. 74524–74538, 2024, <https://doi.org/10.1109/ACCESS.2024.3405978>.
- [18] A. Jafari, J. Faiz, and M. A. Jarrahi, "A Simple and Efficient Current-Based Method for Interturn Fault Detection in BLDC Motors," *IEEE Transactions on Industrial Informatics*, vol. 17, no. 4, pp. 2707–2715, Apr. 2021, <https://doi.org/10.1109/TII.2020.3009867>.